

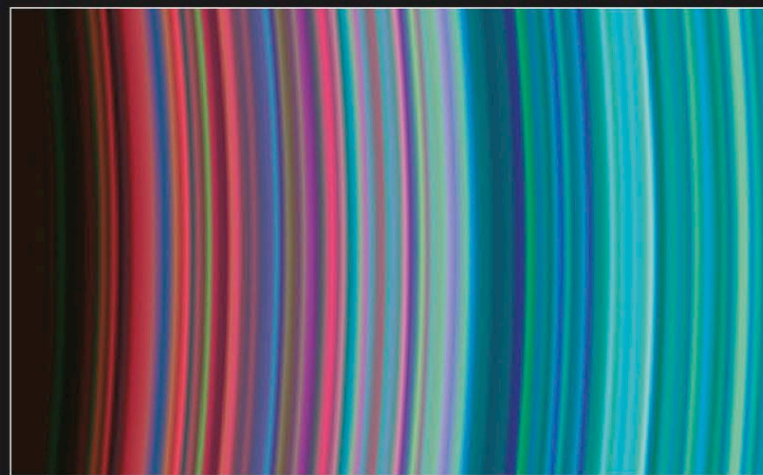
RINGS

Saturn's visible rings are the most extensive, massive, and spectacular in the solar system. From Earth, they appear as a band of material whose appearance changes according to Saturn's position. The rings are, in fact, collections of separate pieces of dirty water ice following individual orbits around Saturn. The pieces range from dust grains to boulders several yards across. They are very reflective, so the rings are bright and easy to see. Individual rings are identified by letters, allocated in order of discovery. The readily seen rings are the C, B, and A rings. These are bounded by others made of tiny particles that are almost transparent. The thin F ring, the broader G ring, and the diffuse E ring lie outside the main rings. The D ring, inside C, completes the system. The rings change slowly over time, and moons orbiting within the system shepherd particles into rings and maintain gaps such as the Encke gap.

MAIN RING SYSTEM

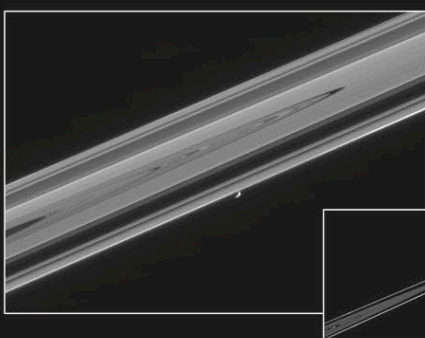
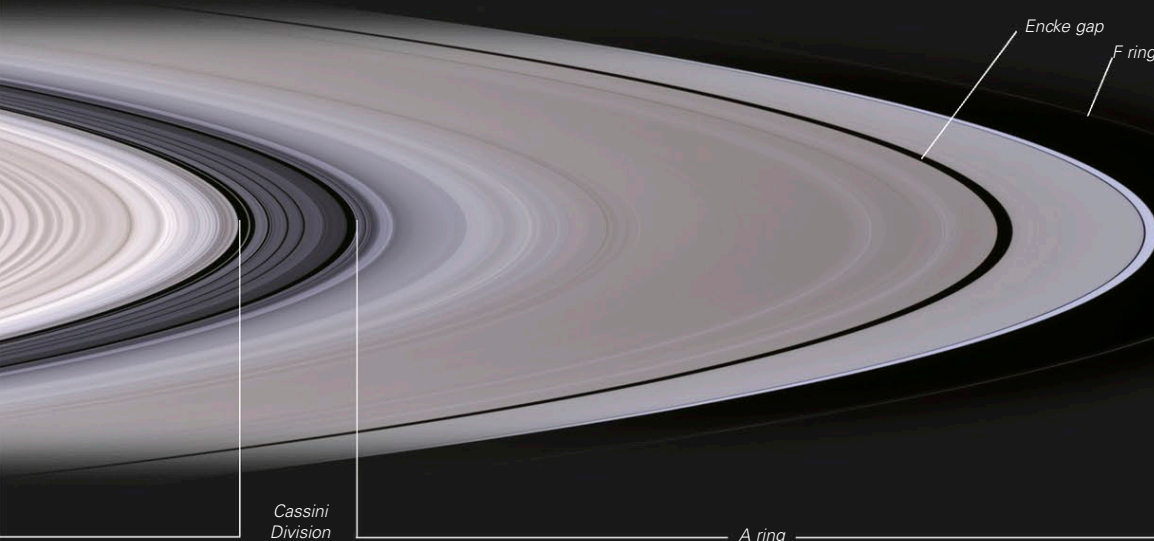
This mosaic of six images shows the main rings in natural color and reveals the ringlets within the Cassini Division. The distance from the inner edge of C to the F ring is about 40,500 miles (65,000 km).

Far beyond the visible system is a huge, doughnut-shaped ring. Almost impossible to see, it was discovered in 2009 by the infrared glow of its cool and sparse dust.



COMPOSITIONAL DIFFERENCES

In this ultraviolet image of the outer portion of the C ring (left) and inner B ring (right), red indicates the presence of dirty particles and turquoise indicates purer ice particles.

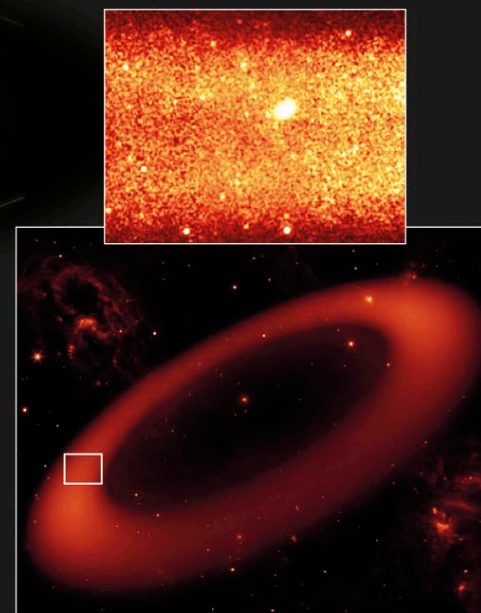


PROMETHEUS AND THE F RING

Saturn's innermost moons orbit within the ring system and interact with it. Some act as shepherd moons, confining particles within specific rings. Prometheus (just below the rings in this image) and Pandora work in this way on either side of the F ring.

PANDORA

The small shepherd moon Pandora orbits just beyond the F ring. It is visible as a white dot in this view taken by Cassini on February 18, 2005.



INVISIBLE RINGS

Saturn's largest ring is invisible to the eye. It consists of dust and was discovered at infrared wavelengths by the Spitzer Space Telescope. Above is an artist's representation of the ring, which lies between about 3.7 and 11.2 million miles (6 million and 18 million km) from Saturn. At the top is a Spitzer image of part of the ring. The ring is tilted about 27 degrees from Saturn's main ring plane.

